

Elbow Muscle Force Optimization: Cost Function n=2 vs n=4

$$\text{Cost function: } u = \left(\sum_i (F_i / PCSA_i)^n \right)^{1/n}$$

Constraints: $\sum M = 0$ (moment equilibrium about elbow), $F_i \geq 0$

Initial conditions: $F_0 = (0, 0, 0)$ for both trials

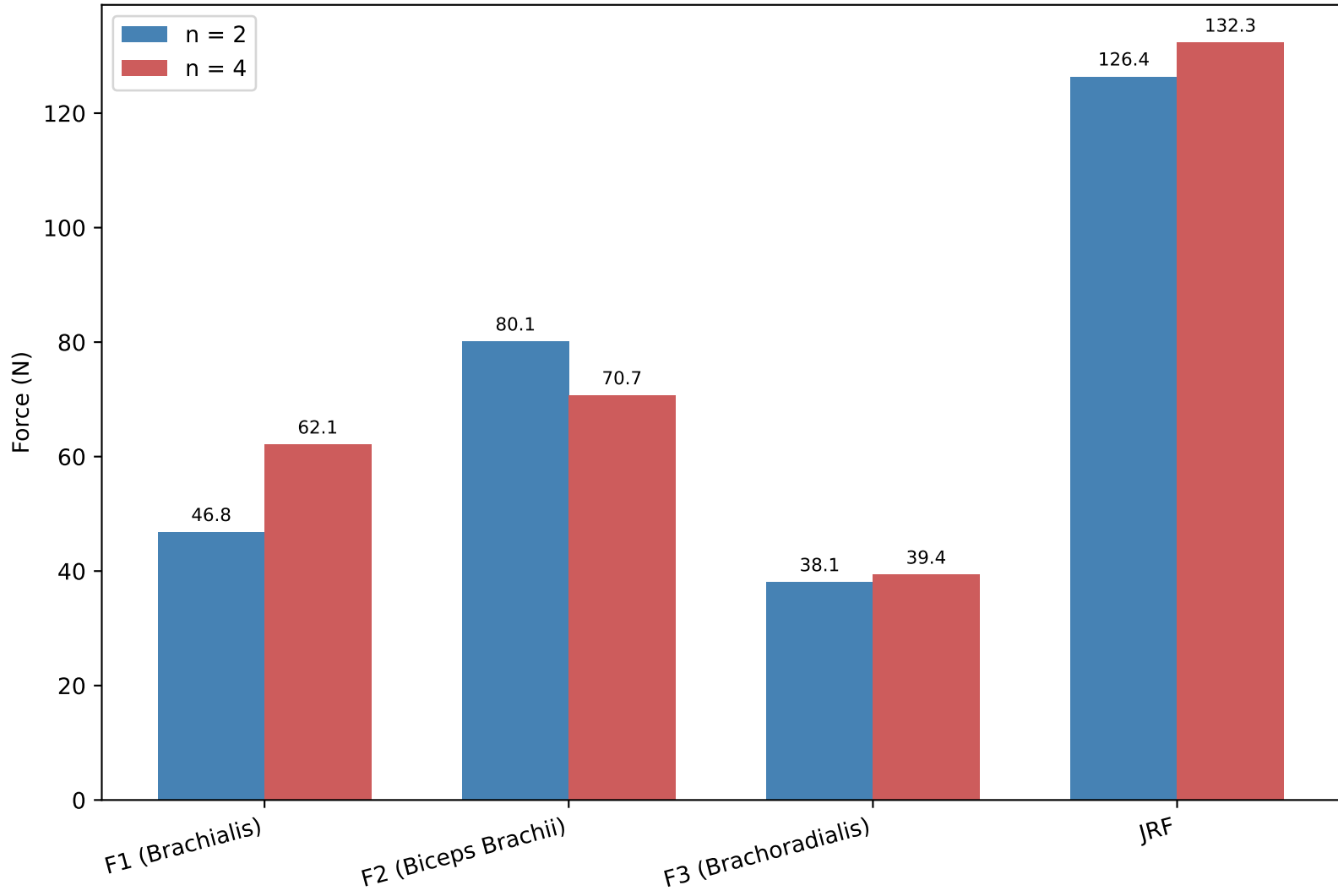
Quantity	n = 2	n = 4	Difference (n4 - n2)
F1 - Brachialis (N)	46.81	62.09	+15.29
F2 - Biceps Brachii (N)	80.10	70.69	-9.41
F3 - Brachoradialis (N)	38.05	39.35	+1.30
JRF x-component (N)	-71.28	-71.94	-0.66
JRF y-component (N)	104.38	111.04	+6.66
JRF magnitude (N)	126.39	132.31	+5.91
JRF angle (deg)	-55.67	-57.06	-1.39
Cost value, u	14.21	11.00	-3.21

Note: u is the n-th-root-normalized (Lp-norm-style) muscle stress, so it is on the same general scale for any n, but n still changes how it weights each muscle's stress (larger n weights the most-stressed muscle more heavily), so u values are only loosely comparable across n -- the forces and JRF are the more meaningful comparison.

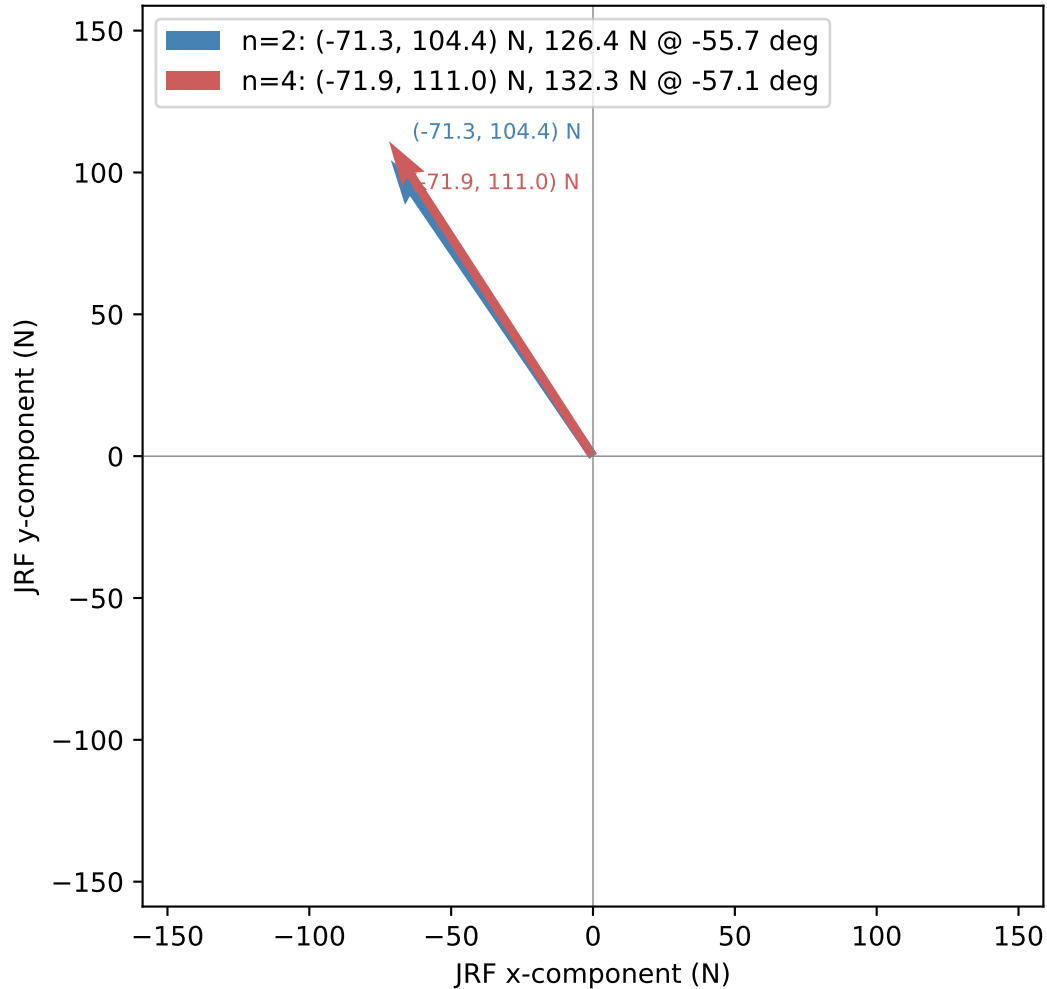
Interpretation

Raising the exponent n from 2 to 4 makes the cost function penalize large individual muscle stresses much more steeply. The n=4 solution redistributes load away from the muscle carrying the largest share under n=2 (Biceps Brachii, F2) and toward the muscle with the smallest moment arm (Brachialis, F1). F3 (Brachoradialis) changes only slightly. The JRF x-component (horizontal, opposing the muscles' combined pull toward the joint) barely changes, while the y-component (vertical, supporting the book and forearm weight) increases more noticeably under n=4, since the larger F1 and F3 pull more steeply than the F2 they partly replace. This yields a slightly larger, steeper-angled resultant JRF under n=4.

Muscle Forces (F1-F3) and Joint Reaction Force: n=2 vs n=4



Joint Reaction Force Vector: n=2 vs n=4



Note: the angle uses $\arctan(j_y/j_x)$ as in the original notebook, which does not account for quadrant -- it will not exactly match the arrow's plotted direction when j_x is negative. Use the (x, y) components above for the true direction.